

Conscious Point Physics: The Hidden Variable Factor $K_0(\lambda)$: Single-CP Path Integral and the Non-Perturbative Nexus Programme

Companion SD#5 — Version 0

Research Agenda Paper — Not a Completed Derivation

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23 March 2026 (agenda recorded); Version 1 target: TBD

STATUS: OPEN PROBLEM AGENDA

**This is a v0 placeholder. The derivation is not yet complete.
This document records exactly what must be done and why it
matters.**

Abstract

SD#4 (Abshier, 2026) completed the CPP superdeterminism series with two exact theorems (macroscopic limit $K \rightarrow 0$; single-CP limit $K = K_0 f_{H_4}$) and one conjecture (the interpolation formula). Two quantitative questions were left open:

1. **The hidden variable factor $K_0(\lambda)$:** What is the explicit functional form of $K_0(\lambda)$, the projection of the particle hidden variable onto the DI-bit exchange degree of freedom?
2. **Non-perturbative proof of the interpolation:** The conjecture $K \approx \varepsilon \cdot K_0 \cdot f_{H_4}$ is supported by a saddle-point calculation but not yet proved to all orders in ε .

These two questions determine the *amplitude* of the CPP correction to Bell correlations. The angular form is already exact (SD#2) and the ratio test $\delta E(36)/\delta E(120) \approx -1.065$ does not depend on the amplitude (SD#2 eq. 21). Nevertheless, a complete theory must provide the amplitude $A_5 = \varphi^{-3}/(2\pi)$ (conjectured) from first principles.

This paper records the mathematical programme required to answer both questions, identifies the specific techniques available from existing CPP results, and states what new mathematics is needed. It is a *research agenda* for future work, not a completed derivation. Upon completion of the derivation, this document becomes Version 1 with the full results.

Keywords: K0 coupling constant, A5 symmetry, golden ratio, Nexus amplitude, research agenda, 600-cell derivation

Plain Language Summary: The final paper in the superdeterminism series outlines the research programme for deriving the Nexus coupling constant K_0 from the 600-cell geometry. This is explicitly labelled as version 0 — a research agenda rather than a completed derivation. The conjectured form involves the golden ratio and the A_5 alternating symmetry group of the icosahedron.

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1 Why This Open Problem Matters

The CPP prediction for Bell test deviations is (SD#4 eq. 9):

$$E_{\text{CPP}}(\theta) = -\cos \theta + \varepsilon \bar{K}_0 f_{H_4}(\theta) + O(\varepsilon^2), \quad (1)$$

where $\varepsilon \approx 2 \times 10^{-26}$ is the Nexus participation fraction (known) and $f_{H_4}(\theta) = A_5 \cos 5\theta + A_3 \cos 3\theta + \dots$ is the angular function from SD#2 (form known, amplitudes A_5, A_3 conjectured).

The factor $\bar{K}_0 = \langle K_0(\lambda) \cos \theta_\lambda \rangle_\lambda$ is the expectation value of K_0 weighted by the hidden variable distribution. Until $K_0(\lambda)$ is derived:

- The *amplitude* of the CPP signal is unknown (but bounded: $|\varepsilon \bar{K}_0| \leq \varepsilon \approx 2 \times 10^{-26}$).
- The conjecture $A_5 = \varphi^{-3}/(2\pi)$ cannot be proved or disproved.
- The non-perturbative verification of the interpolation formula is unavailable.

What is already known without K_0 :

- The angular structure $f_{H_4}(\theta)$ is exact.
- The ratio $\delta E(36)/\delta E(120) \approx -1.065$ is exact (depends only on φ , not on $\bar{K}_0$).
- The null prediction at $\theta = 90$ is exact.
- The scaling laws ($\propto N_q, \propto 1/T$) are exact.

The missing piece is the overall scale, not the shape.

2 The Mathematical Problem

2.1 The single-CP path integral

From SD#4 Theorem 2, the single-CP Nexus path integral is:

$$Z_{\text{CPP}}^{(1,1)}(\lambda, \theta) = \int \mathcal{D}[\phi_P] e^{i(S_P[\phi_P] + S_A[-\phi_P] + S_{PA}[\phi_P, \theta])/\hbar}, \quad (2)$$

where the apparatus field is $\phi_A = -\phi_P$ (constrained), and S_{PA} is the DI-bit exchange coupling between the two CPs.

The $K_0(\lambda)$ factor is:

$$K_0(\lambda) = \frac{Z_{\text{CPP}}^{(1,1)}(\lambda, \theta)}{Z_{\text{QM}}^{(1)}(\lambda) \cdot Z_{\text{QM}}^{(1)}(\text{vac})} - 1 \Big|_{\theta\text{-independent part}}. \quad (3)$$

The θ -independent part is the projection of the full integrand onto the $\cos(5\theta)$ mode (the leading f_{H_4} term); the result is $K_0(\lambda)$.

2.2 What makes this hard

Three technical obstacles:

1. The CPP action $S[\phi]$ at the single-Grid-Point level. The QM series (2040a (Abshier and Grok, 2026a)) derives the Schrödinger equation from Z_{QM} in the coarse-grained limit. The single-Grid-Point action $S[\phi_i]$ at lattice resolution is not yet written in closed form. It is known to involve the SSV field equations (companions C1–C2 (Abshier, 2025)) and the DI-bit density/phase Lagrangian, but the explicit form for a single eCP at a single Grid Point requires extracting the lattice-scale limit of those field equations.

2. The exchange coupling $S_{PA}[\phi_P, \theta]$. This is the interaction term between the particle CP and the apparatus CP. Its angular dependence on θ is what generates the f_{H_4} structure. The derivation requires computing the DI-bit exchange matrix element between two adjacent Grid Points in the 600-cell — a calculation in the adjacency operator formalism developed for the QM series (companion 2040b (Abshier and Grok, 2026b)).

3. The Gaussian integration over ϕ_P . Once $S_P + S_A + S_{PA}$ is written explicitly, the integral over ϕ_P is a Gaussian path integral. Evaluating it gives K_0 as a function of the saddle-point value of ϕ_P , which depends on λ through the initial conditions.

2.3 Candidate approaches

Approach A: Direct lattice calculation. Compute S_{PA} explicitly for two adjacent 600-cell Grid Points in relative orientation θ . Evaluate the path integral in the Gaussian approximation. This gives K_0 perturbatively in the DI-bit exchange coupling constant.

Approach B: Transfer matrix. Express the Nexus path integral as a transfer matrix product along the 600-cell adjacency graph. For $N_{\text{part}} = N_{\text{app}} = 1$, this is a 2×2 matrix problem. The exact result for the single-CP limit follows directly.

Approach C: Symmetry argument. The H_4 symmetry of the 600-cell constrains $K_0(\lambda)$ to transform in a specific irrep. The simplest H_4 -covariant function with zero mean over λ is:

$$K_0(\lambda) \sim \varphi^{-3} \sin(\theta_\lambda), \quad (4)$$

which would give $\bar{K}_0 = \varphi^{-3}/2$ and confirm the conjecture $A_5 = \varphi^{-3}/(2\pi)$. This can be established if the relevant H_4 irrep can be identified from the representation theory of the 600-cell.

Approach C is the most tractable and should be attempted first.

3 The Non-Perturbative Interpolation Problem

3.1 What needs to be proved

SD#4 Conjecture 1 states:

$$K(\lambda, \theta_A, \theta_B) = \varepsilon \cdot K_0(\lambda) \cdot f_{H_4}(\theta_A - \theta_B - \theta_\lambda) + O(\varepsilon^2). \quad (5)$$

The saddle-point calculation establishes the leading term. The non-perturbative proof requires showing that all higher-order terms $c_n \varepsilon^n$ are bounded and that the series converges.

3.2 Candidate approach: transfer matrix

For the general case $N_{\text{part}} = 2$, $N_{\text{app}} \gg 1$, the Nexus constraint distributes over $N_{\text{app}} + N_{\text{rest}}$ Grid Points. The transfer matrix along the 600-cell graph is:

$$T_{ij} = e^{iS_{\text{hop}}(i \rightarrow j)/\hbar}, \quad (6)$$

where S_{hop} is the single-hop DI-bit exchange action. The Nexus partition function is:

$$Z_{\text{CPP}} = \text{Tr}(T) \big|_{\text{constrained}}, \quad (7)$$

where the constraint projects onto the zero-total-DI-bit sector.

For $N_{\text{app}} \gg N_{\text{part}}$, the spectral decomposition of $T^{N_{\text{app}}}$ approaches a uniform distribution over the adjacency eigenvalues (by the Perron–Frobenius theorem on the 600-cell graph). This gives the leading correction $K \sim N_{\text{part}}/N_{\text{app}}$ and the convergence of the geometric series in (5).

This is a rigorous programme; it requires the spectral gap of the 600-cell transfer matrix, which is related to the adjacency eigenvalue gap (already known from the QM/EW series: $\lambda_{\text{max}} = 12$, $\lambda_{\text{next}} = 1 + \varphi \approx 2.618$).

4 Research Programme and Milestones

The following tasks, in order of recommended priority:

Step	Task	Method	Feeds into
1	Identify H_4 irrep for K_0	Representation theory of H_4 (Fulton–Harris (Fulton and Harris, 1991))	A_5 conjecture
2	Compute $S_{PA}(i, j, \theta)$	SSV field equations (C2) + 600-cell adjacency operator	K_0 explicit
3	Evaluate single-CP Gaussian	Transfer matrix on 2-site system	K_0 confirmed
4	Spectral gap of 600-cell T	Use $\lambda = \{12, 1 + \varphi, \varphi - 1, \dots\}$	Convergence proof
5	Non-perturbative proof of conj.	Transfer matrix + Perron–Frobenius	SD#4 Conj. proved
6	Write SD#5 v1	Assemble steps 1–5	Journal submission

4.1 What can be done now

Step 1 (symmetry argument) requires only the representation theory of H_4 , which is in Humphreys (Humphreys, 1990) and Fulton–Harris (Fulton and Harris, 1991). This is pure mathematics requiring no new CPP calculations.

Step 4 (spectral gap) requires only the 600-cell eigenvalues already computed in the QM and EW series. The spectral gap is $\lambda_{\text{max}} - \lambda_{\text{next}} = 12 - (1 + \varphi) = 11 - \varphi \approx 9.382$. This large gap implies rapid convergence of the transfer matrix and supports the conjecture.

Steps 2 and 3 require the single-Grid-Point CPP action, which is the main new mathematical input needed.

5 Connection to the $A_5 = \varphi^{-3}/(2\pi)$ Conjecture

If the H_4 symmetry argument (Step 1) establishes $K_0(\lambda) = \varphi^{-3} \sin(\theta_\lambda)$, then:

$$\bar{K}_0 = \langle K_0(\lambda) \cos \theta_\lambda \rangle_\lambda = \varphi^{-3} \langle \sin \theta_\lambda \cos \theta_\lambda \rangle_\lambda = \frac{\varphi^{-3}}{2}, \quad (8)$$

using $\langle \sin \theta \cos \theta \rangle = 1/2$ for a uniform distribution over θ_λ .

The Fourier coefficient would then be:

$$A_5 = \bar{K}_0/\pi = \frac{\varphi^{-3}}{2\pi} \approx 0.0376, \quad (9)$$

confirming the SD#2 conjecture $A_5 \approx \varphi^{-3}/(2\pi)$ from first principles.

This would be a striking unification: the amplitude of the Bell deviation is the same geometric factor $\varphi^{-3} = V_{\text{subgraph}}/V_{600\text{-cell}}$ that governs the electroweak boson masses (EW #2 ([Abshier and Grok, 2026c](#))). The EW masses and the quantum foundations correction would be two manifestations of the same 600-cell geometric dilution.

This potential unification is the deepest motivation for completing SD#5.

6 Open Problems (Formal Registry)

Open Problem 6.1 (Explicit $K_0(\lambda)$ from H_4 symmetry). *Identify the H_4 irrep in which $K_0(\lambda)$ transforms. Derive the explicit functional form from the representation theory constraint. Show that the conjectured form $K_0(\lambda) \sim \varphi^{-3} \sin \theta_\lambda$ is the unique H_4 -covariant, zero-mean, normalised function consistent with the single-CP path integral structure.*

Open Problem 6.2 (Single-CP CPP action $S[\phi_i]$). *Write the CP action for a single CP at a single Grid Point in the 600-cell lattice explicitly. This requires taking the $\Delta s \rightarrow l_P$ limit of the coarse-grained QM action (companion 2040a ([Abshier and Grok, 2026a](#))) and expressing it in terms of the DI-bit density ρ_i and phase φ_i at one site.*

Open Problem 6.3 (DI-bit exchange matrix element S_{PA}). *Compute the DI-bit exchange coupling $S_{PA}[\phi_P, \theta]$ between two adjacent 600-cell Grid Points in relative orientation θ . This gives the angular dependence of the single-CP path integral and confirms the H_4 structure from first principles.*

Open Problem 6.4 (Non-perturbative proof via transfer matrix). *Apply the transfer matrix method on the 600-cell graph to prove SD#4 Conjecture 1 non-perturbatively. Use the spectral gap $\lambda_{\max} - \lambda_{\text{next}} = 11 - \varphi \approx 9.38$ to establish exponential convergence of the geometric series $\sum_n c_n \varepsilon^n$ in (5).*

7 Version History and Target

Version	Status	Content
v0 (this document)	Agenda only	Problem statement, programme, formal open problems
v1 (target: TBD)	Full paper	Steps 1–5 completed; K_0 derived; A_5 confirmed or revised

This v0 document is deliberately filed in the GitHub repository so that the open problem is formally recorded, the mathematical programme is specified, and future contributors can take it up without repeating the analysis in SD#1–4.

8 Conclusion

The CPP superdeterminism programme (SD#1–4) is complete as a qualitative theory. The angular prediction, the scaling laws, and the ratio test are exact. What remains open is the quantitative amplitude $\bar{K}_0$, which determines the overall scale of the Bell deviation.

The key conjecture — $K_0(\lambda) \sim \varphi^{-3} \sin \theta_\lambda$ and $A_5 = \varphi^{-3}/(2\pi)$ — is accessible via the H_4 representation theory of the 600-cell without new physical input. If confirmed, it would unify the electroweak mass sector (where φ^{-3} is the geometric dilution of the subgraph volume) and the quantum foundations sector (where φ^{-3} sets the Bell deviation amplitude) under a single 600-cell lattice constant.

That unification is the prize that SD#5 v1 should deliver.

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Acknowledgements

The CPP programme is registered at OSF (DOI: <https://doi.org/10.17605/OSF.IO/JXE8D>) and maintained at GitHub (<https://github.com/Hyperphysics-Institute/ CPP>).